

Precept 6 - Solutions

Exercise 1 (Not All Random Variables have names)

Suppose we have a random variable X for which $\mathbb{P}(X > x) = 1 - x^4$, $x \in [0, 1]$. Find the expectation, variance, and density function of X .

Solution:

Let's first calculate the density, for $x \in [0, 1]$:

$$f_X(x) = \frac{d}{dx} \mathbb{P}(X \leq x) = \frac{d}{dx} x^4 = 4x^3.$$

Or alternatively $f_X(x) = 4x^3 1_{[0,1]}(x)$. Now let's calculate the first two moments:

$$\begin{aligned} \mathbb{E}[X] &= \int_{\mathbb{R}} x 4x^3 1_{[0,1]}(x) dx = \int_0^1 x 4x^3 dx = \frac{4}{5}, \\ \mathbb{E}[X^2] &= \int_{\mathbb{R}} x^2 4x^3 1_{[0,1]}(x) dx = \frac{4}{6} = \frac{2}{3}. \end{aligned}$$

Thus, we can get the variance $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 \approx 0.03$.

Exercise 2 (Betting on fedoras)

Having acquired some 309 wisdom, Jessica, together with her ORFriends (all over 21), head to Atlantic city over spring break to have some fun and make some money. The city has been way past its glory days for quite some time now, and as far as the casino business goes could certainly use a breath of fresh air. No wonder our heroes get instantly bored and go to the closest coffee shop to regroup and re-think their spring break plans. As they are sipping their coffee and staring at the forming line, it becomes clear to Jessica that wearing fedoras is a thing in this city. After multitude of observations she realizes that a fedora-wearing person is equally likely to arrive at any point in time and on average there is a fixed number μ of independent fedora-wearers arriving every hour.

- a) Jessica offers the group to play the following game: throw a fair die, and record the time when the first fedora-wearer walks in. For every dot on the die Jessica will receive 10 dollars, and for every minute on the clock, she will pay 1 dollar. Jessica is an honest person, so the game she has offered is perfectly fair. How much time does it take on average for a fedora-wearer to arrive? What is the variance of Jessica's return?

- b) (time-permitting) Let's change the game up a bit: now after recording the outcome of the dice and the arrival time Jessica flips an independent coin, and if it comes up heads the arrival time gets erased, and if tails, the die-outcome gets erased. This new game is assumed to be perfectly fair. In this case, how much time does it take on average for a fedora-wearer to arrive? What is the variance of Jessica's return? Give the cdf of the amount of money that changes hands.

Solution:

First of all let us denote by D the outcome of the die throw, and given the information in the problem we can deduce that the first arrival time of a fedora-wearer is $T \sim \text{Exp}(\mu)$ where μ is the hourly rate of arrival. Let R be Jessica's revenue and X be the total amount of money that changes hands.

- a) In this case we know that $R = 10D - 60T$. The game is fair thus we get

$$\mathbb{E}[R] = 0 \Leftrightarrow \mathbb{E}[T] = \frac{1}{6}\mathbb{E}[D] \Leftrightarrow \frac{1}{\mu} = \frac{3.5}{6} \text{ hours} = 35 \text{ minutes.}$$

Now let's calculate the variance! To do that let's note that $\text{Var}(T) = \frac{1}{\mu^2} = \frac{3.5^2}{6^2}$, and $\text{Var}(D) \approx 2.92$ (justify why):

$$\text{Var}(R) = \text{Var}(10D - 60T) = 100 \text{Var}(D) + 60^2 \text{Var}(T) \approx 292 + 60^2 \frac{3.5^2}{6^2} = 1,517,$$

by independence of D and T .

- b) In this case we get that $R|Heads = 10D$ and $R|Tails = -60T$. By the law of total expectation, we get:

$$\mathbb{P}(Heads)\mathbb{E}[10D] + \mathbb{P}(Tails)\mathbb{E}[-60T] = 0 \Leftrightarrow \frac{1}{\mu} = \frac{3.5}{6} \text{ hours} = 35 \text{ minutes,}$$

as $\mathbb{P}(Heads) = \mathbb{P}(Tails) = \frac{1}{2}$. So the mean is the same as for the last question! Now, let us calculate the variance. For that, we note that:

$$\begin{aligned} \mathbb{E}[R^2] &= \mathbb{E}[R^2|Heads]\mathbb{P}(Heads) + \mathbb{E}[R^2|Tails]\mathbb{P}(Tails) = \frac{1}{2} \cdot 10^2 \mathbb{E}[D^2] + \frac{1}{2} \cdot 60^2 \mathbb{E}[T^2] \\ &= 50(\mathbb{E}[D]^2 + \text{Var}(D)) + 1800(\mathbb{E}[T]^2 + \text{Var}(T)) \approx 759 + 1800 \cdot 2 \cdot \frac{3.5^2}{6^2} = 1,984. \end{aligned}$$

Since $\mathbb{E}[R] = 0$, we get $\text{Var}(R) = \mathbb{E}[R^2] - \mathbb{E}[R]^2 = \mathbb{E}[R^2] \approx 1,984$. Note that the variance in this case is larger than in the previous case even though we have the same exact parameters and expectations!

Now note that $X = |R|$ and thus its cdf will have the following form:

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x|Heads)\mathbb{P}(Heads) + \mathbb{P}(X \leq x|Tails)\mathbb{P}(Tails) \\ &= \frac{1}{2}\mathbb{P}(D \leq \frac{x}{10}) + \frac{1}{2}\mathbb{P}(T \leq \frac{x}{60}) = \begin{cases} \frac{1}{12}\lfloor \frac{x}{10} \rfloor + \frac{1}{2}(1 - e^{\mu x/60}), & \text{if } x \in [0, 60) \\ \frac{1}{2} + \frac{1}{2}(1 - e^{\mu x/60}), & \text{if } x \in [60, \infty). \end{cases} \end{aligned}$$

Exercise 3 (Forest Fire)

The number of forest fires in California in a given year is a random variable. This number depends in part on the length of the drought season. Assume that the length of the drought season X is exponential with parameter μ . Given $X = x$ assume that the number of fires is conditionally Poisson with mean x . Find the probability that there are at least three fires this season.

Solution:

Denote the number of forest fires by N , and thus the problem asks us to calculate $\mathbb{P}(N \geq 3)$. The information given in the problem states the conditional distribution $N|X = x \sim \text{Poisson}(x)$. From here using the law of total probability we get:

$$\begin{aligned}\mathbb{P}(N \geq 3) &= \int_0^\infty \mathbb{P}(N \geq 3|X = x)\mathbb{P}(X \in dx) = 1 - \int_0^\infty \mathbb{P}(N < 3|X = x)\mathbb{P}(X \in dx) \\ &= 1 - \int_0^\infty \left(\frac{e^{-x}x^0}{0!} + \frac{e^{-x}x^1}{1!} + \frac{e^{-x}x^2}{2!} \right) \mu e^{-\mu x} dx \\ &= 1 - \mu \left(\int_0^\infty e^{-(\mu+1)x} dx + \int_0^\infty e^{-(\mu+1)x} x dx + \int_0^\infty \frac{1}{2} e^{-(\mu+1)x} x^2 dx \right) \\ &= 1 - \frac{\mu}{\mu+1} - \frac{\mu}{(\mu+1)^2} - \frac{\mu}{(\mu+1)^3},\end{aligned}$$

where we used that the density of $\text{Gamma}(k, \mu+1)$ integrates to 1, here for $k = 1, 2, 3$ (see Example 3.1.7 of the lecture notes).

Exercise 4 (Fish Delivery)

Jerry has decided to open a new sushi restaurant "Prince-Udon" in Princeton, and since there are no affordable fish farms in the local area he decides to get his supplies from the fish market in Philly. Fortunately for him, a new app, FoodEx, provides ingredient delivery services for small restaurant owners. Before getting to Jerry the fish will travel to a distribution center A in between Princeton and Philly. Being a meticulous entrepreneur, Jerry wants to know how much time it takes for the fish to get from Philly to the distribution center and from the distribution center to Princeton. FoodEx obliges and tells Jerry that the joint density of Philly to A travel time X and the total travel time Y is given by

$$\mathbb{P}(X \in dx, Y \in dy) = f_{X,Y}(x, y) dx dy = \lambda^2 e^{-\lambda y} dx dy$$

for $0 \leq x \leq y < \infty$. What are the marginal densities of the total and Philly to A travel times? Find the conditional distribution of total travel time given that the travel time from Philly to A is x . Find the distribution of the travel time from A to Princeton.

Solution:

We start off by computing the marginal density of X . For $0 \leq x < \infty$

$$\begin{aligned}
\mathbb{P}(X \in dx) &= \int_{y=x}^{\infty} \mathbb{P}(X \in dx, Y \in dy) \\
&= \int_{y=x}^{\infty} \lambda^2 e^{-\lambda y} dy dx \\
&= \left[\frac{\lambda^2 e^{-\lambda y}}{-\lambda} \right]_{y=x}^{\infty} dx \\
&= \lambda e^{-\lambda x} dx
\end{aligned}$$

Hence, $X \sim \text{Exponential}(\lambda)$. We operate similarly for Y . For $0 \leq y < \infty$,

$$\begin{aligned}
\mathbb{P}(Y \in dy) &= \int_{x=0}^y \mathbb{P}(X \in dx, Y \in dy) \\
&= \int_{x=0}^y \lambda^2 e^{-\lambda y} dx dy \\
&= \lambda^2 y e^{-\lambda y} dy \\
&= \frac{(\lambda y)^{2-1} \lambda e^{-\lambda y}}{(2-1)!} dy
\end{aligned}$$

Hence, $Y \sim \text{Gamma}(2, \lambda)$.

The conditional density of Y given $X = x$ is

$$\mathbb{P}(Y \in dy | X = x) = \frac{\mathbb{P}(Y \in dy, X \in dx)}{\mathbb{P}(X \in dx)} = \lambda e^{-\lambda(y-x)} dy,$$

for $y \geq x$ and 0 otherwise. We used the marginal distribution of X found above.

Now let us denote the travel time from A to Princeton by $W = Y - X$:

$$\begin{aligned}
\mathbb{P}(W \leq w) &= \mathbb{P}(Y - X \leq w) \\
&= \int_0^{\infty} \mathbb{P}(Y - X \leq w | X = x) \mathbb{P}(X \in dx) \\
&= \int_0^{\infty} \mathbb{P}(Y \leq w + x | X = x) \mathbb{P}(X \in dx) \\
&= \int_0^{\infty} \int_x^{w+x} \mathbb{P}(Y \in dy | X = x) \mathbb{P}(X \in dx) \\
&= \int_0^{\infty} \left(\int_x^{w+x} \lambda e^{-\lambda(y-x)} dy \right) \lambda e^{-\lambda x} dx \\
&= \int_0^{\infty} (1 - e^{-\lambda w}) \lambda e^{-\lambda x} dx \\
&= 1 - e^{-\lambda w},
\end{aligned}$$

for $w \geq 0$. Hence, $W \sim \text{Exponential}(\lambda)$. For the fourth equality, we used that the conditional density of Y given X is non-zero only when $y \geq x$.